

DFI 5.2 Port → MC Block Map

DDR5, one channel. Every DFI port: which MC block drives it (out) / receives it (in), and status. Dir: out = MC→PHY, in = PHY→MC.

Widths parametric (N_RANKS, DQ, GEAR). _pN = per-phase (×GEAR).

done = walked + figured **stub** = slot exists, detail later **to-do** = not started

1 Command interface (out unless noted)

Note: cmd_type on the CA bus (via DFI_CMD_ENCODE) covers *all* CA opcodes — demand (ACT/RD/WR/PRE) *and* maintenance/power (REF/REFsb/RFM/M). The ME injects these into the bundle, so power/maint *commands* share the command phase. Only the pure CKE-level transitions (power-down PDE/PDX, and CKE-low during self-refresh) ride dfi_cke (POWER_MGMT), not a CA opcode. So power management uses BOTH: a CA command (encoder) + the CKE pin.

DFI port	Dir	Width	MC block	Status	Note
dfi_address_pN	out	14×GEAR	DFI_CMD_ENCODE	done	fig_70; CA-encoded 2-UI
dfi_cs_pN	out	N_RANKS×GEAR	DFI_CMD_ENCODE / CS_GEN	done	fig_70; one-cold active-low
dfi_reset_n_pN	out	1	INIT_FSM	done	init sequence
dfi_cke_pN	out	N_RANKS×GEAR	POWER_MGMT	stub	power-down / self-refresh
dfi_odt_pN	out	N_RANKS×GEAR	WL data-timing / POWER_MGMT	to-do	RTT_WR window (WL-offset)
dfi_dram_clk_disable_pN	out	per-clk	POWER_MGMT	stub	clock-stop
dfi_2n_mode_pN	out	1	INIT / cfg	stub	config-set
dfi_cs_geardown_pN	out	1	INIT / cfg	stub	DDR5 geardown
dfi_parity_in_pN	out	1	DFI_CMD_ENCODE (parity)	to-do	CA parity gen
dfi_alert_n_pN	in	1	ALERT/ERR mon	stub	CA-parity / write-CRC retry

2 Write data interface (out)

DFI port	Dir	Width	MC block	Status	Note
dfi_wrddata_pN	out	2·DQ×GEAR	WR_SPLIT	done	fig_68/73; from WD_SRAM
dfi_wrddata_en_pN	out	slices×GEAR	WL_LAUNCH	done	tphy_wrlat shift line
dfi_wrddata_mask_pN	out	2·DQ/8×GEAR	WR_ACCUM / WR_SPLIT	done	DM / narrow flag
dfi_wrddata_cs_pN	out	PHY_RANK×GEAR	WL_LAUNCH	to-do	data-cs (rank timing)
dfi_wrddata_crc_pN	out	DATA/8×GEAR	WR CRC gen	stub	link CRC
dfi_wrddata_ecc_pN	out	opt	WR ECC gen	stub	if controller ECC

3 Read data interface

DFI port	Dir	Width	MC block	Status	Note
dfi_rddata_wN	in	2·DQ×GEAR	RD_ACCUM	done	fig_68/73; into RD_SRAM
dfi_rddata_en_pN	out	slices×GEAR	RD_CAP	done	tphy_rldat shift line
dfi_rddata_valid_wN	in	slices×GEAR	RD_ACCUM	done	gates capture
dfi_rddata_cs_pN	out	PHY_RANK×GEAR	RD_CAP	to-do	data-cs (rank timing)
dfi_rddata_dnv_wN	in	slices×GEAR	RD_ACCUM	to-do	data-not-valid
dfi_rddata_dbi_wN	in	DATA/8×GEAR	RD path	stub	read DBI
dfi_rddata_crc_wN	in	DATA/8×GEAR	RD CRC check	stub	read link CRC

4 Status / Update (Control-clk, single-phase)

DFI port	Dir	Width	MC block	Status	Note
dfi_init_start	out	1	INIT / FREQ FSM	done	init / freq-change start
dfi_init_complete	in	1	INIT / FREQ FSM	done	gates DFI_MUX
dfi_cmd_freq_ratio	out	2	FREQ_CHANGE FSM	stub	= GEAR select
dfi_data_freq_ratio	out	2	FREQ_CHANGE FSM	stub	
dfi_freq_fsp	out	1-2	FREQ_CHANGE FSM	stub	MR set-point
dfi_frequency	out	W	FREQ_CHANGE FSM	stub	target freq code
dfi_ctrlupd_req	out	1	UPDATE ctrl	to-do	MC-requested update
dfi_ctrlupd_ack	in	1	UPDATE ctrl	to-do	
dfi_phyupd_req	in	1	UPDATE ctrl	to-do	PHY-requested update
dfi_phyupd_ack	out	1	UPDATE ctrl	to-do	
dfi_phyupd_type	in	2	UPDATE ctrl	to-do	

5 Error / Low-power (Control-clk)

DFI port	Dir	Width	MC block	Status	Note
dfi_error	in	1	ALERT/ERR mon	stub	PHY error strobe
dfi_error_info	in	W×4	ALERT/ERR mon	stub	error source
dfi_lp_ctrl_req	out	1	POWER_MGMT	to-do	LP control handshake
dfi_lp_ctrl_ack	in	1	POWER_MGMT	to-do	
dfi_lp_ctrl_wakeup	out	1	POWER_MGMT	to-do	

DFI port	Dir	Width	MC block	Status	Note
dfi_lp_data_req	out	1	POWER_MGMT	to-do	LP data handshake
dfi_lp_data_ack	in	1	POWER_MGMT	to-do	
dfi_lp_data_wakeup	out	1	POWER_MGMT	to-do	

6 Block status summary

MC block	status	drives / receives
DFI_CMD_ENCODE	done	dfi_address, dfi_cs (+parity to-do)
WL_LAUNCH / WR_SPLIT	done	dfi_wrdata, _en, _mask (cs/crc pending)
RD_CAP / RD_ACCUM	done	dfi_rddata_en, rddata, valid (cs/dnv pending)
INIT_FSM	done	dfi_reset_n, dfi_init_start/complete
DFI_MUX	done	CA driver select (init vs sched)
POWER_MGMT	stub	dfi_cke, dfi_odt, dram_clk_disable, lp_*
FREQ_CHANGE FSM	stub	dfi_*_freq_ratio, freq_fsp, frequency
UPDATE ctrl	to-do	dfi_ctrlupd_*, dfi_phyupd_*
ALERT/ERR mon	stub	dfi_alert_n, dfi_error, error_info
CRC / ECC / DBI	stub	wrdata_crc/ecc, rddata_crc/dbi

Data-path core (command encode + write/read data) is **done**. Remaining are maintenance/power/config/error side-paths (**stub/to-do**) — slots defined, detail pending. All per-phase (_pN) signals are xGEAR wide; the output-register stage after DFI_MUX drives them onto the DFI pins in **dfi_clk**.